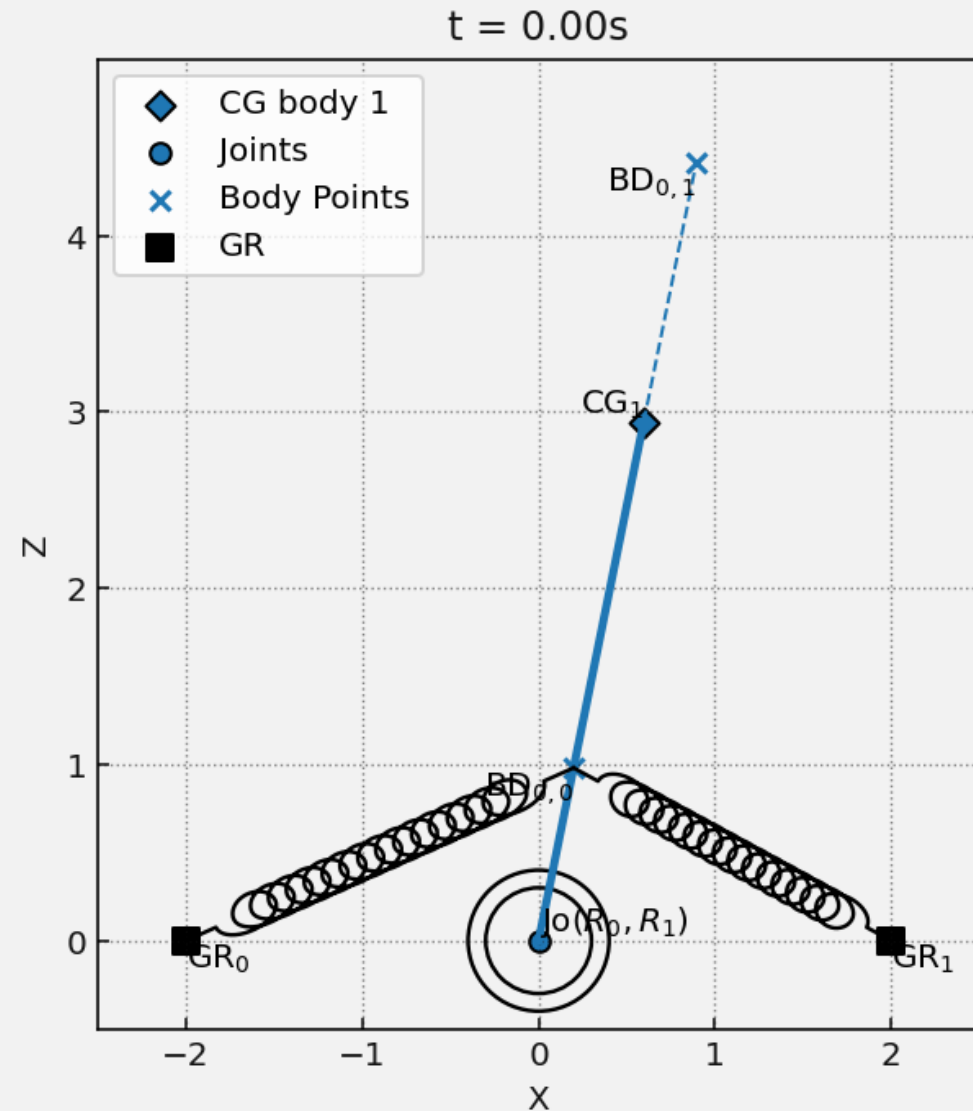


FLAP symbolic

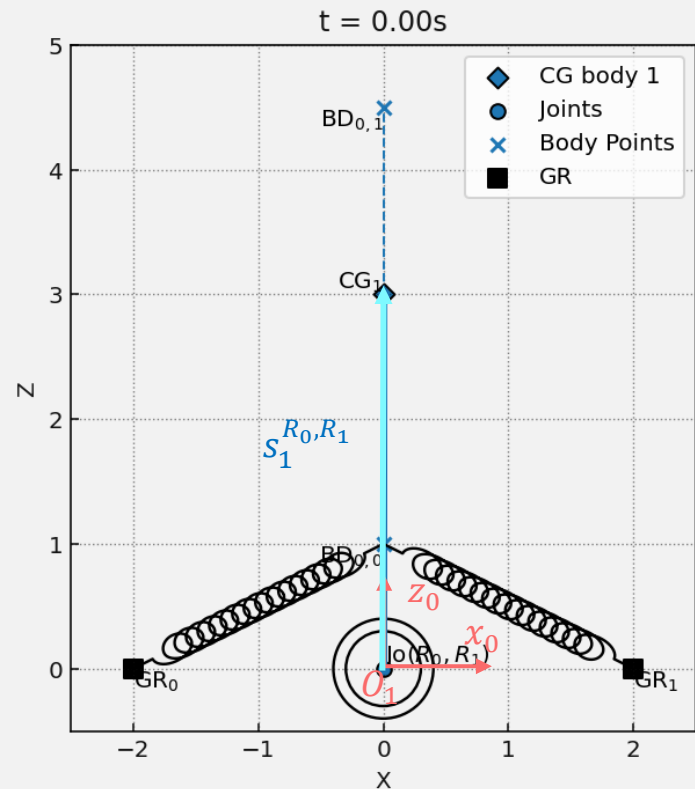
(previous example with symbolic parameters)



Bodies definition: symbolic flap's length, L

Define a list of strings ['L'] which are transformed into variables (sympy symbol)

```
dataNames = ['L']
BodyDataSym = symVars_definition(dataNames, globals())
```



Connecting bodies

Joint type

```
joints = [[0, 1]]
types = ['R']
parent_cg_to_joint = [[0, 0]]
joint_to_child_cg = [[0, L/2]]
prismatic_direction = [[np.nan]]
```

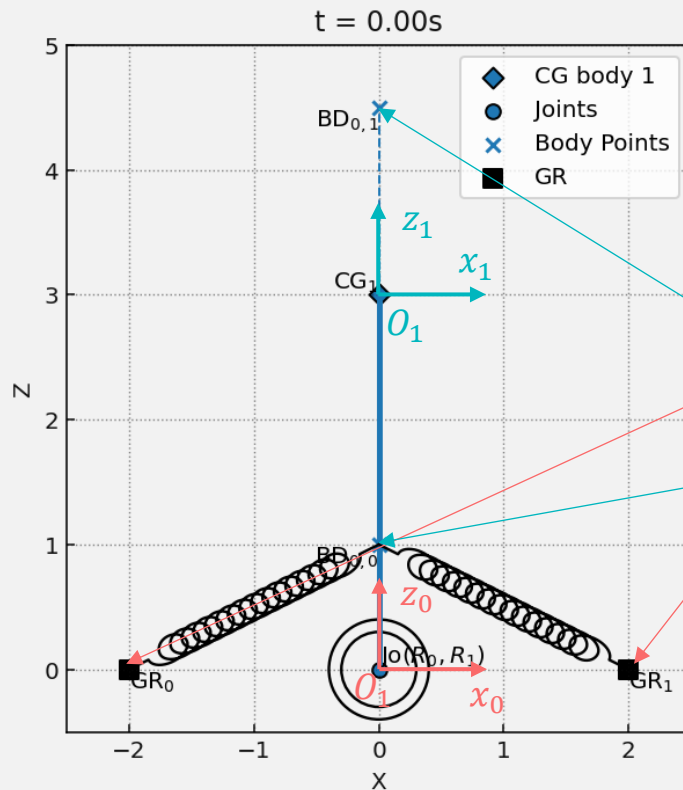
Direction of allowed translation

Go to next slide to continue animation

Points definition:

Define a list of strings ['a', 'b'], which are transformed into variables (sympy symbol)

```
PointNames = ['a', 'b']
PointsSym = symVars_definition(PointNames, globals())
```



Note: vectors measured from the appropriate origin (see colors)

```
Initial_Points["GR"] = [[-b, 0], [b, 0]]
```

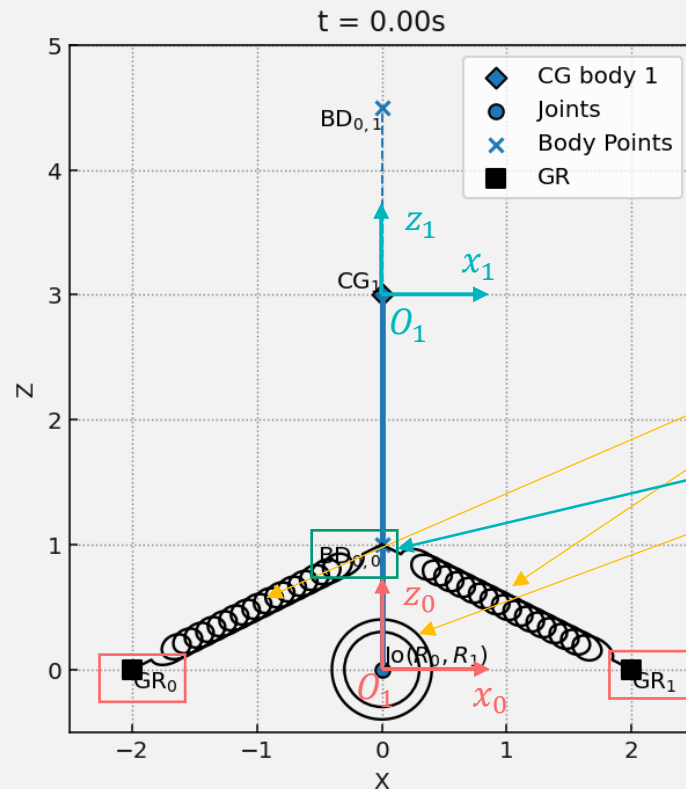
```
Initial_Points["BD"] = {}
Initial_Points["BD"][1] = [[0, a-L/2], [0, L/2]]
```

If not specified, the symbolic variables are given the value of 1 on evaluation.

Go to next slide to continue animation

Forces definition: symbolic stiffness, k

Springs and dampers are defined as a list of tuples



Force["TensionSpring"] = [((('BD10', 'GR00'), [2.2361, k]),
((('BD10', 'GR01'), [2.2361, k])),
]

Force["TorsionSpring"] = [((0, 1), [0, 100])]]

Undeformed
length, l_0

Stiffness, k

Undeformed
angle, θ_0

Stiffness, k_θ

Go to next slide to continue animation